

Chip Implementation

(Relevant to projects 1, 2, 3, 5)

Files: For each chip Xxx we provide a skeletal Xxx.hdl program, also called *stub file*, with a missing PARTS section. In addition, for each chip we provide an Xxx.tst script that tells the hardware simulator how to test the chip, along with an Xxx.cmp compare file containing the correct outputs that the supplied test is expected to generate. Your task is writing, and testing, the chip implementations (specifically: Completing the supplied Xxx.hdl files).

Contract: For each chip in the list, your chip implementation (modified Xxx.hdl file), tested by the supplied Xxx.tst file, must generate the outputs listed in the supplied Xxx.cmp file. If the actual outputs generated by your chip disagree with the desired outputs, the simulator will report error messages.

Tips

0. Before implementing a chip, it is recommended to experiment with its builtin implementation. If you are using the online simulator, simply click the *builtin* toggle; If you are using the legacy desktop simulator, load the builtin chip from the tools/builtInChips folder.
1. Each chip can be implemented in more than one way. The simpler the implementation, the better. As a general rule, strive to use as few chip-parts as possible.
2. Although any chip can be implemented directly from Nand gates only, you can use any other chip from the current or previous projects as a chip-part, as needed (see the previous tip).
3. There is no need to build “helper chips” of your own design. Your HDL programs should use only the chips described in the course.
4. We recommend implementing the chips in the order in which they are listed. If, for some reason, you don’t complete the HDL implementation of some chip, you can still use it as a chip-part in other HDL programs. In the online simulator, the chip evaluation process uses builtin chip implementations, so there is no further ado. If you are using the legacy desktop version, rename the chip file that you did not implement, or remove it from the project folder. This will force the desktop simulator to use the chip’s builtin implementation.

Each chip can be tested interactively, or using a test script. In the online simulator, simply run the test script. In the desktop simulator, you must first load the chip’s test script, and then run it.

Tools

[If you are using the Nand2Tetris IDE Online](#) (which is recommended), all the Xxx.hdl, Xxx.tst and Xxx.cmp files are available in your browser memory. To develop and test a particular chip, select the project / chip from the simulator’s drop-down menus. Your edited HDL code will be saved automatically. To download the HDL files to your local PC, click the *download* button. The current version of all the project’s Xxx.hdl files will be downloaded as one zip file.

Alternatively, using the legacy desktop simulator is also possible. If you've downloaded the Nand to Tetris software suite from www.nand2tetris.org, and extracted it into a folder named nand2tetris on your computer, the nand2tetris/tools folder contains the desktop version of the hardware simulator, and the nand2tetris/projects/*n* folder contains all the files needed for completing project *n*. You can write/edit the HDL code of each Xxx.hdl file using any plain text editor, and then test your code using the desktop simulator.

Documentation comment: We sometimes use abbreviated notation. For example, the HDL comment “if (in) out = 1, else out = 0” means: “if (in = 1) then set out to 0, else set out to 1”. Likewise, the condition “if (a and b) ...” means “if (a = 1 and b = 1) ...”. And so on.

References

[HDL Guide](#)

[Chips Set API](#)

Tutorials

The tutorials listed below were written for the legacy desktop version of the hardware simulator. However, the working principles illustrated in these tutorials can be used to perform similar actions in the online simulator (a major difference is that the online simulator requires no file loading or saving – everything is stored in your browser memory).

[Hardware Simulator: Intro](#)

[Building and Testing Chips](#)

[Script-Based Chip Simulation](#)

[Hardware Simulator Tutorial](#) (click *slideshow*)

Consult each reference / tutorial as needed: There is no need to go through the entire resource.